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A Reflective Journey: Navigating Your Cumulative Experience at Iowa State University

When I started at Iowa State University in the spring of 2023 as a Cybersecurity Engineering student, I thought most of my experience would be about learning programming, cybersecurity, and computers. Now I understand that engineering is also about teamwork, communication, problem solving, ethics, and learning how to adapt when something does not work, change the perspective of thinking.

One of my favorite classes has been CYBE 3310, Application of Cryptographic Concepts to Cyber Security. I liked this class, because I could finally see how some of the cybersecurity topics I learned before are actually used in the real world, such as learning about cryptography helped me understand why things like authentication, encryption, keys, and secure communication are important. It also made me more interested in cybersecurity, because I could see how these topics are used to protect real information.

One of the hardest classes I have taken was CPRE 2880, Embedded Systems 1. The labs could take around 10 to 15 hours every week. One challenge was finding enough time to work with my lab partner because we both had different schedules. This class taught me that even if I understand the technical part of something, I still need enough time to test it, fix problems, and work with other people.

The final project in CPRE 2880 was also challenging because our group became a team of five people. Before that, I was mostly working with one other person. With five people, we had to

divide the work differently and trust everyone to complete their part. During some of the demos, only a few people could be involved directly while others controlled the CyBot. We lost some points during the demos, but I learned that teamwork and communication can be just as important as the technical work.

We also had to use the C and Python coding languages in CPRE 2880. Most of the class was based on C, but we used Python for the GUI, where we had many problems trying to get the map and object detection to work correctly. We had to keep testing things and figuring out what was wrong. This helped me become more comfortable with troubleshooting. Now, when something does not work the first time, I try to break the problem into smaller parts instead of immediately thinking that everything is wrong.

COMS 3090 was another important class for my learning experience. As my team created an Android application, and I worked mainly on the front end and the design of the app. I was also a communication leader for the group. I liked how our team worked together and helped each other troubleshooting's the issue. We had people focusing on the back end and others focusing on the front end, so communication between us was very important.

One problem we had was with WebSockets. The people working on the back end were setting up the WebSocket using the login and authentication process. On the front end, I was first trying to connect without doing it the same way. Because of that, the connection was not working correctly, so I had to understand how the back end was doing it and change my part to match it. This taught me that sometimes I have to change my approach instead of continuing to do something the same way and communication, while creation is really crucial. It also showed me that different parts of a software project have to work together.

CYBE 2340, Legal, Professional, and Ethical Issues in Cyber Systems, also changed the way I think about cybersecurity. Before this class, I thought more about the technical side of cybersecurity. The class helped me understand that cybersecurity also includes laws, privacy, ethics, and professional responsibility. Knowing how an attack works does not mean that someone should use that knowledge without permission. Cybersecurity should be used to protect systems and people.

I have also seen this at my job with ISUF. One time, I helped troubleshoot problems with logging in and accessing shared files. I had to think about why the person could not access the files while also understanding that permissions are there to protect information. The goal was not to find a way around security. The goal was to make sure the correct person had the correct access. This helped me connect what I learn in class to a real work environment.

A lot of my learning also happens outside of class. I ask many questions during office hours and talk to TAs and professors when I do not understand something. I used to worry that some of my questions were too simple, but I learned that it is better to ask than to stay confused. I am also involved with Hackathon Club and Hackers Club. These clubs help me learn about cybersecurity topics outside of normal classwork. I also like following cybersecurity news so I can see what is happening in the real world.

My Senior Design project is called LLM Agentic System for Circuit Card Design. Our goal is to create an AI based system that can help automate some parts of the early circuit card design process. We are working with things like Python, Linux, Docker, cloud servers, large language models, and circuit card design. Some of these topics are new to me because my major is focused more on cybersecurity. Because of this project, I have to learn things that are outside of

what I already know.

Senior Design has also taught me about scheduling and teamwork. We have several meetings every week, and everyone has different classes and responsibilities. Sometimes it is difficult to find enough time for everything. It has made me realize that working on a real engineering project is not only about the technical work. You also need to communicate, organize your time, and learn from people who have more experience.

If I could start college again, I would probably keep most things the same. Even the difficult classes helped me learn something. However, I would spend more time learning topics beyond what was required for an assignment. Sometimes I was focused mainly on finishing the lab or meeting the deadline. I would also try to meet more people earlier and manage my schedule better.

In the future, I am interested in becoming a cybersecurity analyst or working with cybersecurity and data. I want to work with systems, find security problems, understand threats, and help protect information. I know that cybersecurity changes very quickly, so graduating will not mean that I am finished learning. I will have to continue learning new tools, technologies, attacks, and ways to protect systems.

Overall, Iowa State has helped me grow both technically and personally. I have learned programming, cybersecurity, cryptography, software development, embedded systems, and many other technical skills. At the same time, I have learned how important teamwork, communication, ethics, time management, and problem solving are.

The biggest thing I have learned is that I do not need to know the answer to everything right away. I need to know how to ask questions, look for information, work with other people, and

keep trying when something does not work. I think these skills will help me as I finish my degree and continue into a career in cybersecurity.